

SmartBite: Food Recognition & Nutrition Estimator

Project in Machine Learning and Data in Operations

Spring 2025

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GitHub: github.com/SimonLuder/SmartBite



Project Goal

- **End-to-end ML Repository**

- Train new classifiers
- Classification of food images
- Provide nutrition information via API and a interactive Dashboard

Welcome to SmartBite! 🍴🍷

Your AI-Powered Meal Analysis Assistant

How it works:

1. Upload a photo of your meal.



Data

- **Dataset**

- Food-101 dataset
- 101 classes (different food types)
- 1000 images per class (max. 512 x 512 pixels)

- **Processing**

- Accessible from remote S3-bucket on Digitalocean using dvc
- Pre-processing via Python script

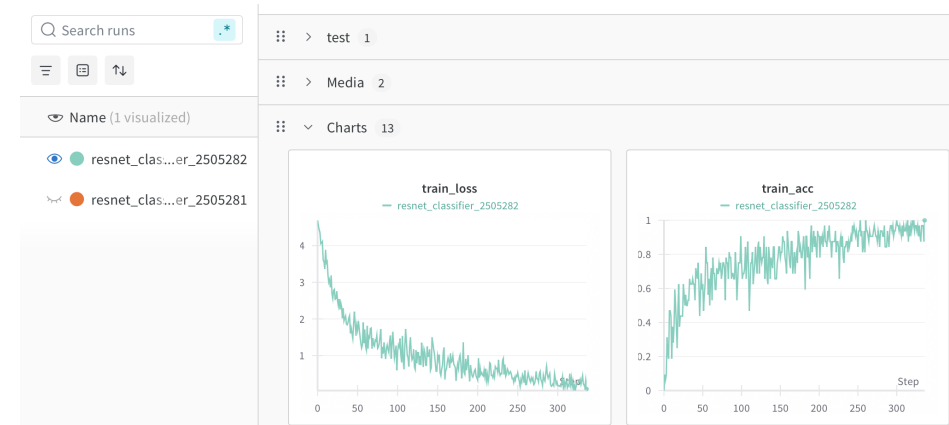


Example image: Crème brûlée

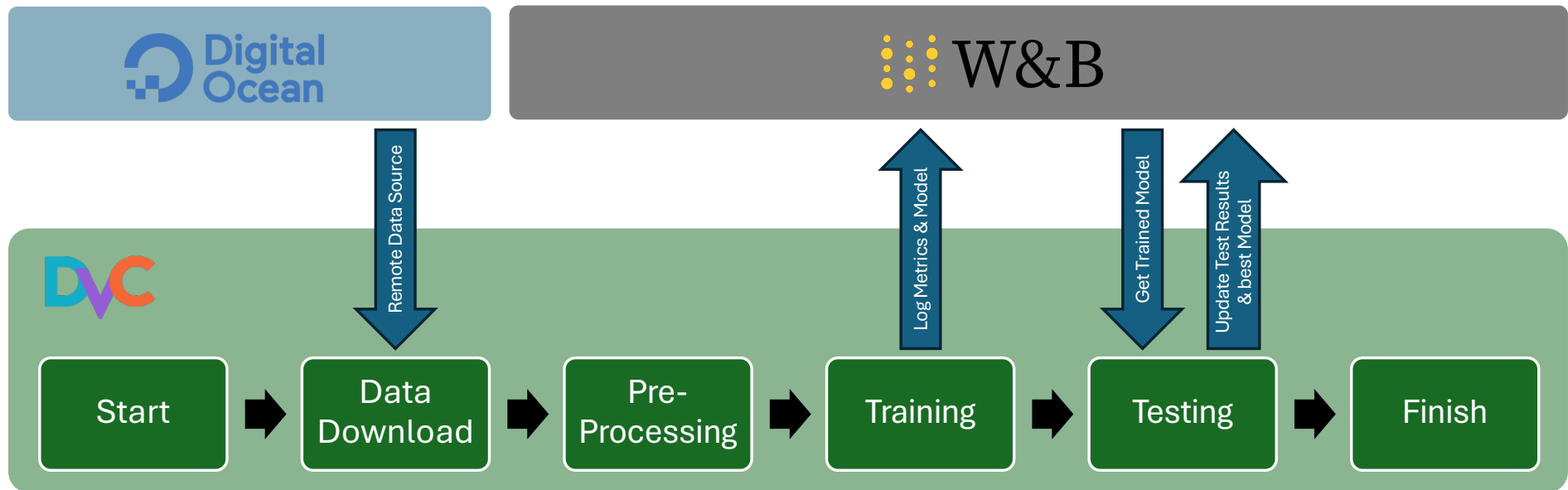
Model Training and Logging



- **Purpose**
 - Train the model to classify images using pytorch lightning
 - Log the metrics and loss for comparison of the models and training runs
- **Why we use pytorch lightning**
 - Already written training loops
 - Simpler modular code
 - Lot of additional functionalities (logging)
- **Why we use WandB**
 - Online hosting for sharing between different members
 - Tracks loss, metrics, model



Modelling Pipeline



- Run with “dvc repro”
- Executable in docker container

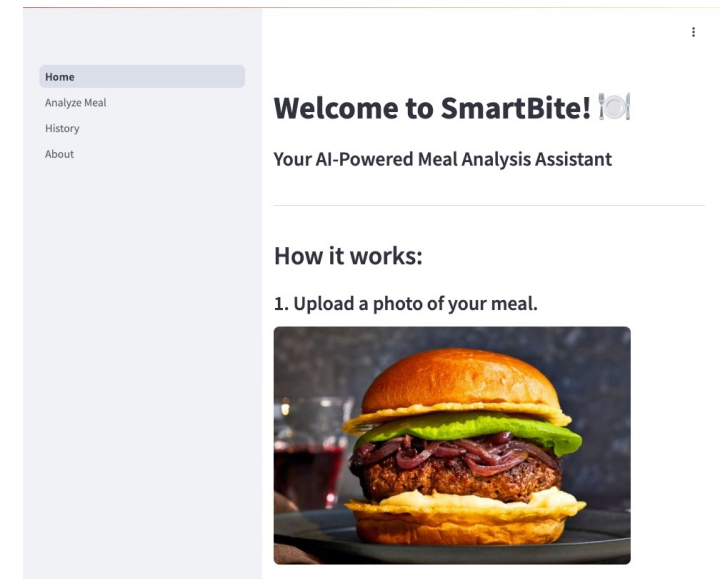
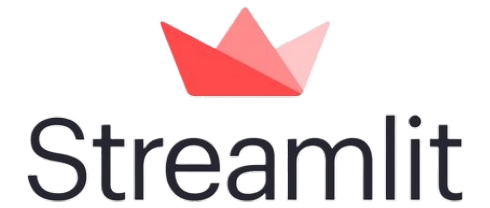
Frontend

- **Purpose:**

- Provides a simple and interactive interface
- Collects user input (images)
- Sends input to backend for prediction
- Displays model results (food class & nutrition info)
- Stores analysis history in sessionStorage

- **Why we use Streamlit**

- Simple and intuitive
- Integrates seamlessly with ML workflows
- Lightweight and fast development



Backend

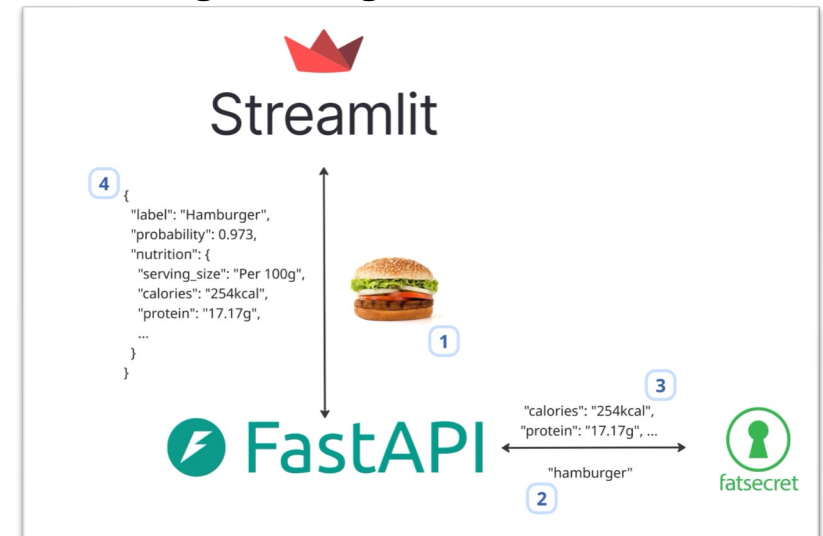
- **Purpose**

- Exposes REST API endpoints
- Instantiates and loads the model
- Classifies uploaded food images
- Fetches nutrition information
- Communicates with the Streamlit frontend

- **Why we use FastAPI**

- Flexibility & Control
- Async API
- Lightweight & Deployment-Friendly

From Image to Insight



CI/CD Workflow with GitHub Actions



tests

Download
Model
Weights

Build backend
Docker Image

Run
Integration
Tests

Upload Model
Weights

build-and-push

Download
Model Weights
Artefact

Build backend
and frontend
Docker Image

Push Images to
GHCR

deployment



Triggered on:

- Push to main
- Pull requests
- Manual run

Demo



<https://smartbite-frontend.onrender.com>

